

Lab Template

1. Fill in the following table. Be sure to include at least 2 sentences describing what each storage system is. Also, include when each type of storage should be used and whether or not it was used/accessed/showcased in the lab.

(10 points)

Name	Description	Usecase
Block	In this case, we store the data in “blocks” and each block has its own address and the OS also has the rights to read and write directly to any block.	It is basically used in the places where data needs to be accessed quickly. In this lab, we used it when we were viewing partitions such as /dev/sda1 and /dev/sda2 with lsblk, which are block devices.
Object	Over here, we basically store the data in discrete units called as “objects” which contain data, metadata (it basically tells about the object’s properties, content or usage. For example if we have stored a photo in the object storage then it will tell us about its filename, owner, the date when the file was uploaded etc.) and a unique identifier(It basically helps in identifying the object easily as each object has a unique identifier.). We can access it through web APIs.	It is used in cloud storage systems where large scale backups or unstructured data is required. We haven’t used it in this lab since all storage was local and could be accessed directly.
File	This basically organizes the data that is stored in various files and folders, where each file and folder has a name and path that tells the system where to find it, and a file system	It is used for storing files that we make on a daily basis. We used it in this lab when we mounted /dev/sda2 and accessed Windows system in /mnt/Windows/System32/c

	basically manages how these files are supposed to be saved, read and organized on the disk.	onfig.
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2. Explain what the -o, rw, and uid are for in the following command

```
mount -t ntfs-3g -o rw,uid=1000 /dev/sda2 /mnt
```

(10 points)

Ans) In this command, -o provides us with the options that help us in customizing the way we should open the drive, rw tells us that we can take a look at the files and edit the information they contain, and uid gives us permission to the user so that he can open and edit the files.

3. How is the Security Account Manager (SAM) file similar to/different from /etc/shadow? Include information about the locations of the file, information that is stored in each, who has access to the files, and the format the data is stored in.
(20 points)

Ans) The SAM file is located in the C:\Windows\System32\Config\SAM and it stores usernames, security identifiers and password hashes for all the local windows accounts. Over here, the passwords can be encrypted using the NTLM hashing algorithm and since the SAM can only be accessed by the Windows operating system itself, the administrators cannot open or edit it while windows is running and the file is locked. Moreover, the file is not human readable (since its a binary file) and it requires special tools like chntpw.

The /etc/shadow file is located in the /etc/ directory on the linux systems and stores the usernames and hashed passwords, password expiration dates and their account policy. Over here, only the root user has the permission to read or edit this file. Moreover, it is a plaintext file, which follows a specific format separated by colons.

Over here, we can see that both the files have the same purpose, that is, protect the data for user authentication, however, they differ in location, format and access control. Over here, only the SAM file is binary and is basically locked by the system, whereas /etc/shadow is a plaintext file and is protected by strict permissions.

4. Rerun “chntpw -l SAM” and take a screenshot of the “Blank password” field to include in your lab submission
(10 points)

Ans)

```
(root@kali)-[/mnt/Windows/System32/config]
# chntpw -l SAM
chntpw version 1.00 140201, (c) Petter N Hagen
Hive <SAM> name (from header): <\SystemRoot\System32\Config\SAM>
ROOT KEY at offset: 0x001020 * Subkey indexing type is: 686c <lh>
File size 65536 [10000] bytes, containing 7 pages (+ 1 headerpage)
Used for data: 355/32120 blocks/bytes, unused: 16/8616 blocks/bytes.
```

RID	Username	Admin?	Lock?
01f4	Administrator	ADMIN	
03ef	Alex		
01f7	DefaultAccount		dis/lock
03e8	defaultuser0		dis/lock
01f5	Guest		
03ec	James		
03ed	Lilly		
03ee	Seregil		
03e9	yellowsnow	ADMIN	*BLANK*

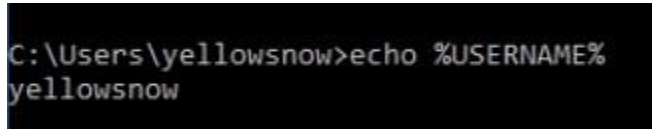
5. What if you would have found the hash `aad3b435b51404eeaad3b435b51404ee` in the LM hash portion of your file? What does it mean? Especially, if there are multiple instances of that same hash value.
(10 points)

Ans) The hash `aad3b435b51404eeaad3b435b51404ee` in the LM portion of the file indicates that the file is an empty string or has a null password, which means that the file is highly insecure and if we find multiple instances of the same hash, it means that there are several users who have a null password, which tells us that it is a major security vulnerability, since it becomes easy to hack their systems.

6. Take a screenshot running the following command to prove that you've successfully logged in as the yellowsnow user:

```
echo %USERNAME%
```

(10 points)

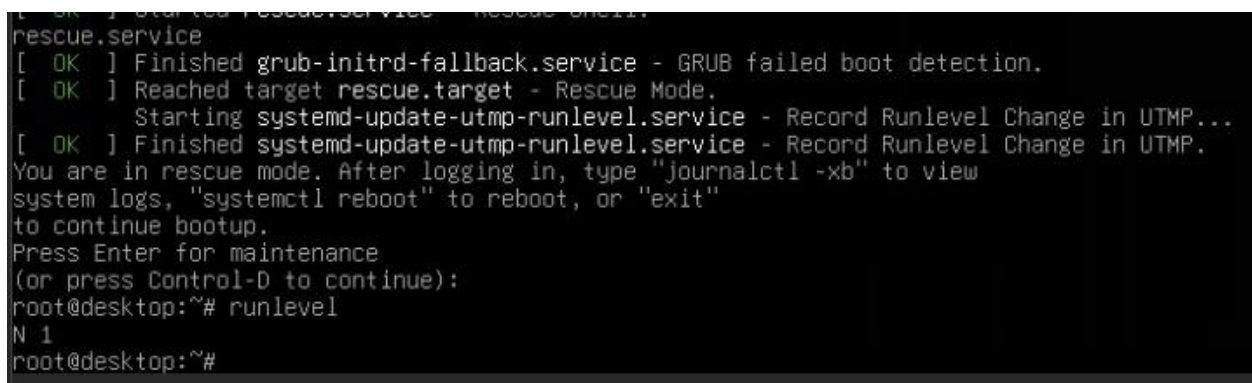


```
C:\Users\yellowsnow>echo %USERNAME%
yellowsnow
```

7. Take a screenshot of the following command to verify that you are in single-user mode (should return N 1):

```
runlevel
```

(10 points)



```
[ OK ] Started Rescue.service - Rescue shell.
rescue.service
[ OK ] Finished grub-initrd-fallback.service - GRUB failed boot detection.
[ OK ] Reached target rescue.target - Rescue Mode.
Starting systemd-update-utmp-runlevel.service - Record Runlevel Change in UTMP...
[ OK ] Finished systemd-update-utmp-runlevel.service - Record Runlevel Change in UTMP.
You are in rescue mode. After logging in, type "journalctl -xb" to view
system logs, "systemctl reboot" to reboot, or "exit"
to continue bootup.
Press Enter for maintenance
(or press Control-D to continue):
root@desktop:~# runlevel
N 1
root@desktop:~#
```

8. Pick two of the following controls: Secure Boot, BitLocker (Windows), LUKS (Linux full-disk encryption), making USB ports non-bootable / disabling external-boot in firmware. Describe how each works (1–2 sentences) and how it would prevent or reduce the risk demonstrated in today's lab. (20 points)

Ans) 1) Secure boot: It is a feature in BIOS that basically allows the system to boot operating systems that are trusted by the manufacturer of the computer. It helps us in preventing risks as it stops attackers from booting to a live Linux image such as Kali. In the lab, it would have prevented us from loading the Kali live environment to access and modify the Windows disk offline.

2) BitLocker: It encrypts the entire hard drive using strong encryption algorithms, which makes all data unreadable without using the password and the encryption key. It can prevent the risk, as even if someone is logged in or removes the drive to read it(as we did while mounting /dev/sda2), the attacker can only see the encrypted data, as it completely blocks the attack that was demonstrated, where we mounted and browsed Windows files using Kali.